

Glitches & Hazards

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Glitches & Hazards

- ♦ When the input to a combinational circuit changes, unwanted switching transients (Glitches) may appear in the output.
- ♦ These transients occur when different paths from input to output have different propagation delays.
- ♦ A hazard is said to exist when a circuit has the possibility of producing such a glitch.

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Types of Hazards

Static 1 - Hazard

- ♦ Possibility of a circuit's output producing a 0 glitch when we would expect the output to remain at a steady 1.



Static 0 - Hazard

- ♦ Possibility of a circuit's output producing a 1 glitch when we would expect the output to remain at a steady 0.



Dynamic Hazard

- ♦ Possibility of the output of a circuit changing more number of times when it should change from (0 to 1) or (1 to 0).



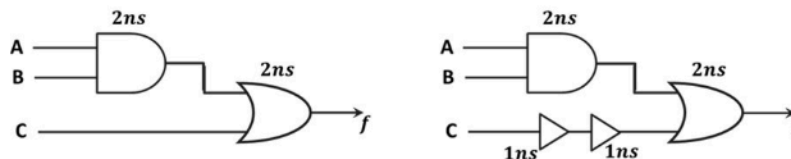
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Elimination of Hazards

Technique 1

- ♦ Adding delay to balance the delays in the parallel paths can eliminate hazards since, unequal delays is the main cause of these glitches.



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Elimination of Hazards

Technique 2

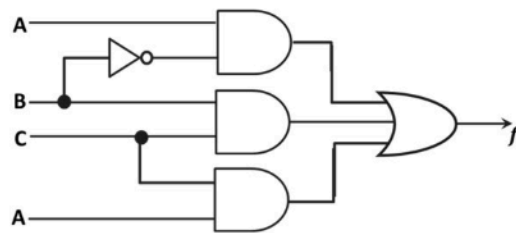
- ♦ Adding the redundant gates can eliminate the hazards.

$$f = \sum m(3,4,5,7)$$

		BC			
A		00	01	11	10
	0	0	0	1	0
	1	1	1	1	0

$$f = AB' + BC + AC$$

$$f = A\bar{B} + BC + AC$$



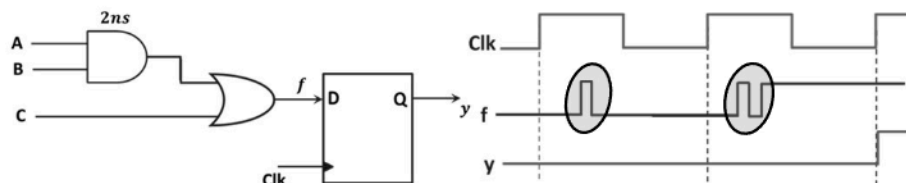
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Elimination of Hazards

Technique 3

- ♦ Adding the pipelining registers at the outputs the combinational circuits.



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Memories

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Agenda

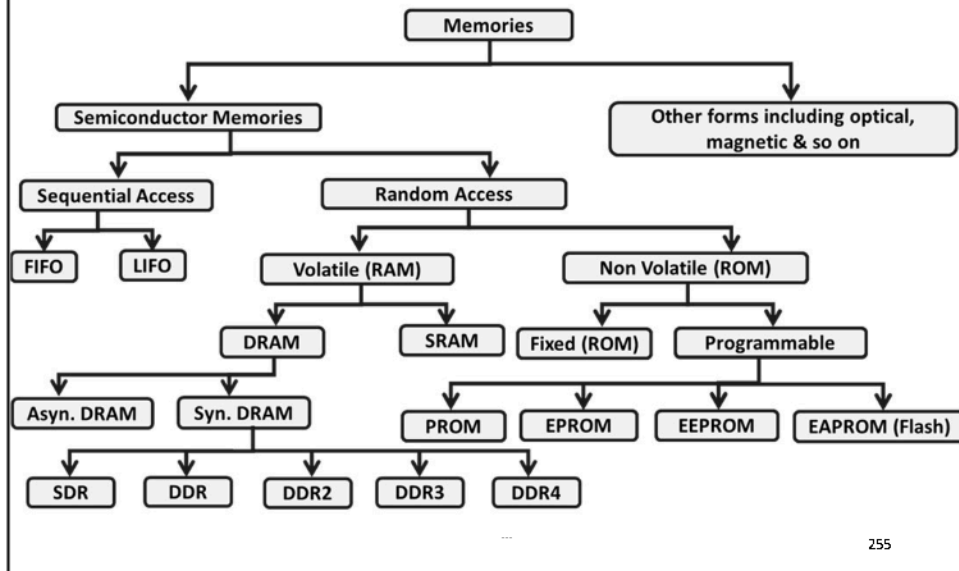
- ◆ Classification of Memories
- ◆ Basics of Memories
- ◆ Composition of Memories

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Classification of Memories



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Classification of Memories

- ♦ Optical memories like CD, DVD and magnetic memories like floppy, HDD will come under other forms of memories.
- ♦ Let's concentrate more on semiconductor memories.
- ♦ **Sequential Access Memories** : In these memories, data cannot be accessed from random address locations. It can be accessed in a serial manner.
- ♦ **Random Access Memories** : In these memories, data can be accessed from random address locations.
- ♦ FIFO & LIFO comes under sequential access memories.
- ♦ **Volatile Memories** : In these memories, stored data will be lost, if there is a loss in power supply.

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Classification of Memories

- ♦ **Non - Volatile Memories** : In these memories, stored data will not be lost, even if there is a loss in power supply.
- ♦ SRAM & DRAM will come under volatile random access memories.
- ♦ **Synchronous DRAM** : In these memories, the basic read & write will be done in synchronism with a clock signal.
- ♦ **Asynchronous DRAM** : In these memories, the basic read & write will not be done in synchronism with a clock signal.
- ♦ **Single Data Rate**: 1 Byte of data for each clock cycle, either at posedge or negedge.
- ♦ **Double Data Rate** : 2 Bytes of data for each clock cycle. One data at posedge & second data at negedge.

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Classification of Memories

- ♦ **Double Data Rate – 2nd Gen** : 4 Bytes of data for each clock cycle. two bytes at posedge & two more bytes at negedge.
- ♦ **Double Data Rate – 3rd Gen** : 8 Bytes of data for each clock cycle. four bytes at posedge & four more bytes at negedge.
- ♦ **Double Data Rate – 4th Gen** : 16 Bytes of data for each clock cycle. eight bytes at posedge & eight more bytes at negedge.
- ♦ In these days, DDR4 is the commonly used memory in LAPTOP's & Servers.
- ♦ **ROM**: Data stored in these memories is fixed at the time of fabricating the memories and cannot be changed again.
- ♦ No further write operation is possible. Only READ operation is possible.
- ♦ **Programmable ROM**: These memory chips can be programmed only once.

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Classification of Memories

- ♦ **Erasable Programmable ROM** : These memory chips can be programmed for any no. of times & data will be erased by the application of UV light rays.
- ♦ Data stored in these memories can be programmed by the user based on his requirement. Both read & write are possible in these memories.
- ♦ **Electrically Erasable Programmable ROM**. These memory chips can be programmed for any no. of times & data will be erased by the application of electrical signals.
- ♦ **Erasable Alterable Programmable ROM**. These memories are special type of EEPROMs and data can be written at the block level instead of byte level.
- ♦ Can be programmed while in the circuit. USB drives, Memory cards & SSD's will come under this category.

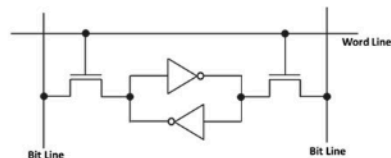
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SRAM vs DRAM

Static RAM

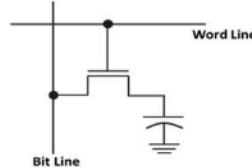
- ♦ Static RAM uses static devices like transistors & MOSFET's to store the data.



- ♦ At least, 6 no. of transistors are required for implementing one bit memory cell.
- ♦ Not area efficient. Occupies more area.
- ♦ No periodic refresh required.
- ♦ Less access time b/c of static devices.
- ♦ Standard IC fabrication process is sufficient for fabricating these memories.

Dynamic RAM

- ♦ Dynamic RAM uses dynamic devices like capacitors to store the data.



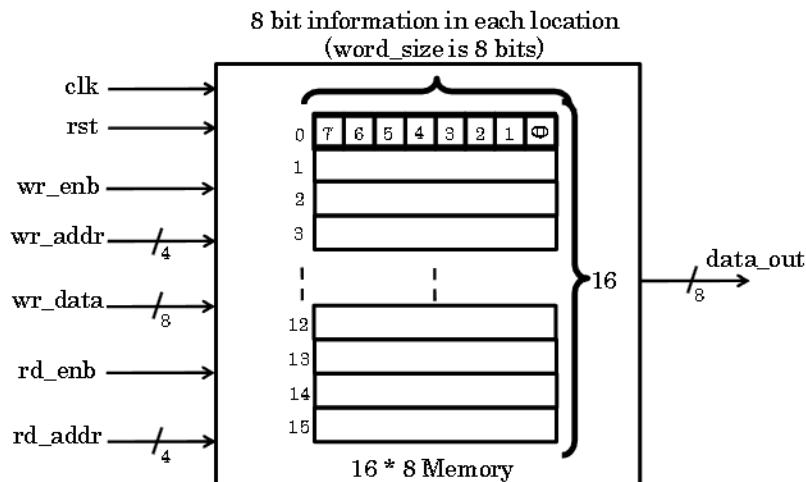
- ♦ One transistor & one capacitor are sufficient for implementing one bit memory.
- ♦ Area efficient. Occupies less area.
- ♦ Periodic refresh required.
- ♦ High access time b/c of usage of capacitors.
- ♦ Special IC fabrication process is required for fabricating these memories.

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Basic Concepts of Memories



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Composition of Memories

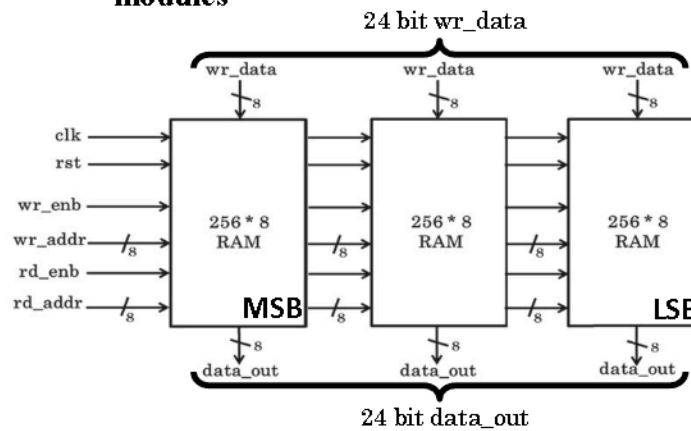
- ♦ It is the process of creating large memory module using smaller memory module.
- ♦ The following are the different requirements.
 - ♦ We may need to construct a large memory with larger word size (more bits per word) or
 - ♦ We may need to construct large memory with Wider addressing space (more words) or
 - ♦ Both

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Memory with larger word size

Example : Construct 256*24 RAM using 256*8 RAM modules



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Memory with more words

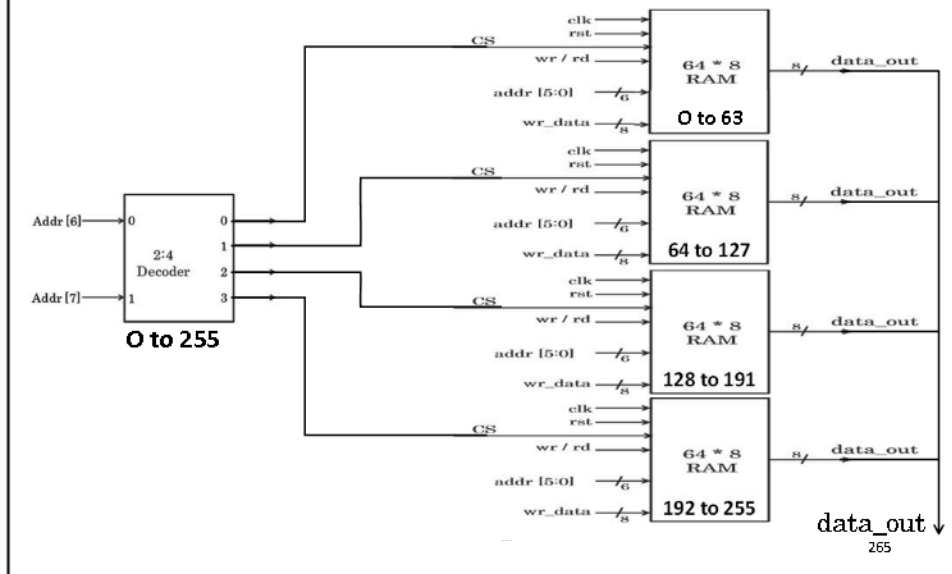
Example : Construct 256*8 RAM using 64*8 RAM modules

- ♦ For constructing 256 * 8 Memory using 64 * 8 Memories, we need 4 modules of 64 * 8 memories since $256/64 = 4$.
- ♦ The available 64 * 8 Memory will have 6 address lines.
- ♦ But, the required memory should have 8 address lines.
- ♦ The extra 2 lines can be used for selecting any of the 4 smaller memory modules by using a decoder logic.
- ♦ Recall the fact that, only one of the output line of a decoder will be at HIGH level depending on the input available.

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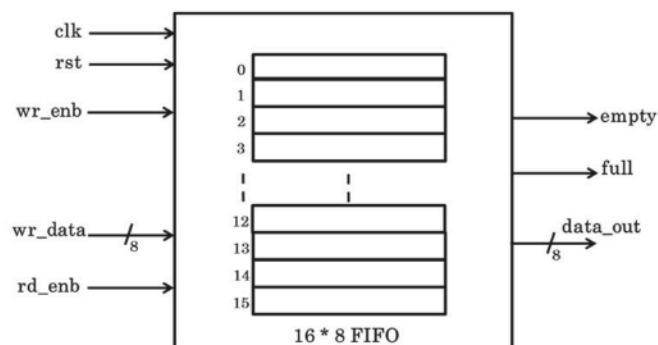
Memory with more words



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FIFO

- ♦ Used to pass data between two asynchronous clock domains.
- ♦ Used to pass data between two systems working with same clock but with different throughput.



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Cache Memory

Speed



Data Storage Capacity



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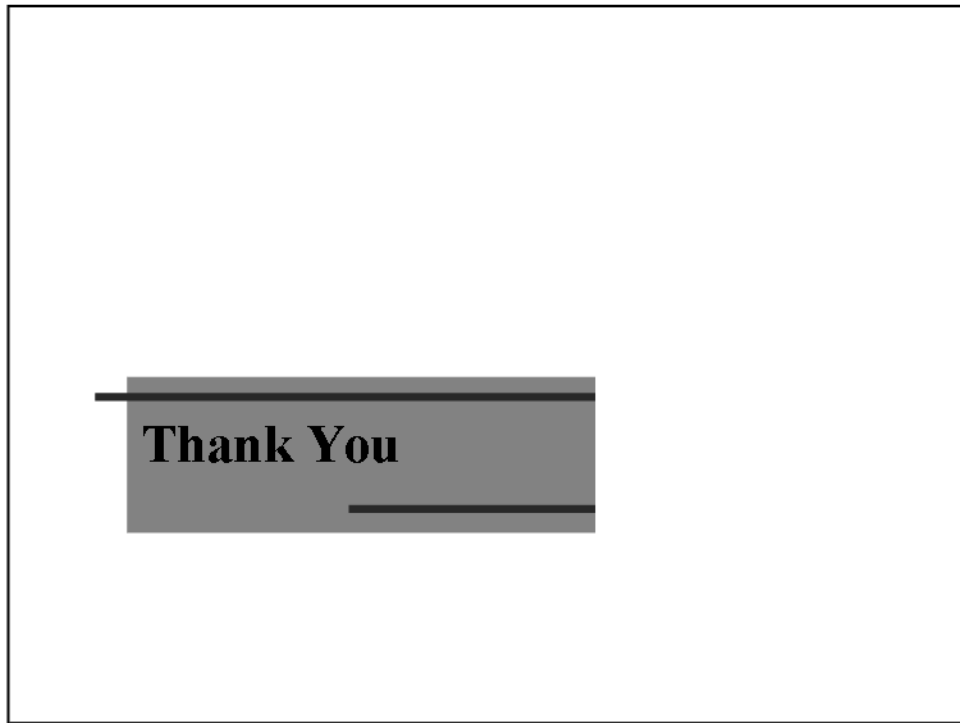
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Cache Memory

- ♦ CPU Contains Registers and Cache Memory So, that it can access frequently accessed data at a faster speed instead of accessing from slower memories like RAM & HDD.
- ♦ CPU has only few MB of Cache. But, CPU makes a tremendous use of this small memory because the speed of data access in cache is high because it need not do periodic refresh unlike DRAM.
- ♦ Cache uses SRAM Technology. So, Many no. of memory cells cannot be fabricated in a small processor. So, cache memories will be in the order of MB unlike DRAM which will be in the order of GB.
- ♦ For a better performance, Use a processor with more Cache Memory.

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